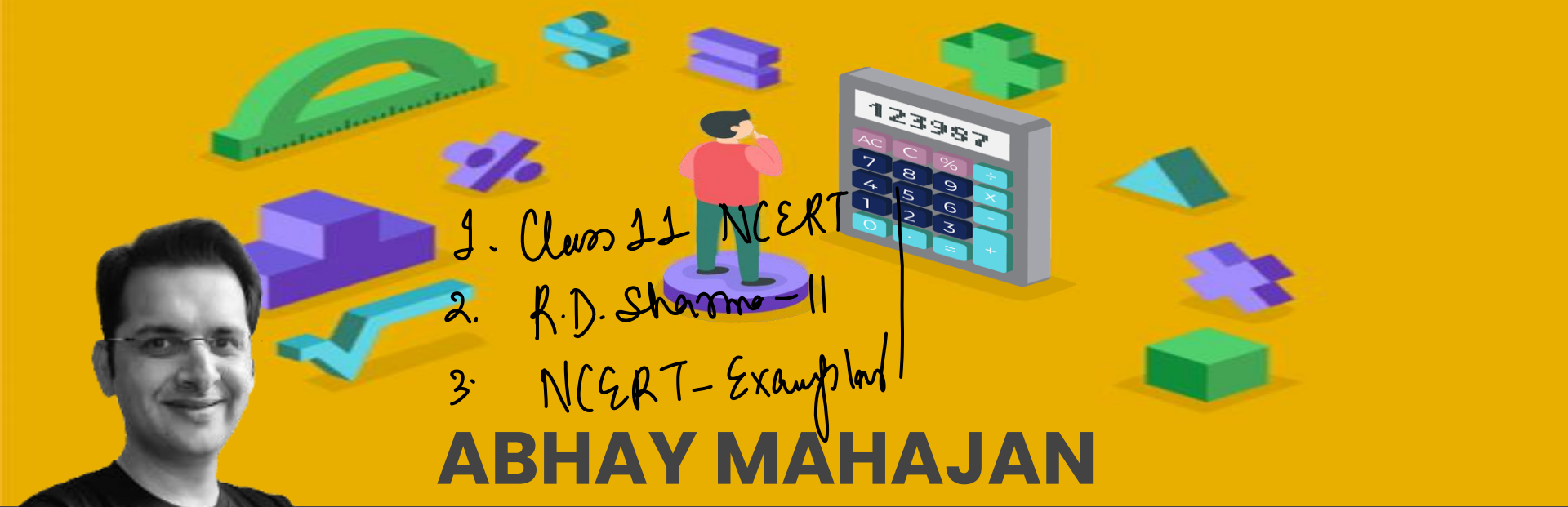




- 
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B.Tech. IIT Roorkee

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**ABHAY
MAHAJAN**

Number Theory

**MONDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Kinematics and Laws of Motion:
Most Important Questions**

**WEDNESDAY
@ 12:30 PM**

**ASIF
TYAGI**

**Animal Tissue:
Part-1**

**WEDNESDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

Atomic Structure

**FRIDAY
@ 11:30 AM**

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Solutions and Colligative Properties

**SATURDAY
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MATHEMATICS EXPERT



Content Creator :

Sahil Jindal

B.Tech, NIT Kurukshetra

4 Year JEE Teaching
Experience.

Trigonometry !!

Sine Rule and Cosine Rule



270, -90

Let's define $\cos\theta$ and $\sin\theta$.

Consider a unit circle centered at the origin.

Draw any angle θ .

X coordinate of P is defined as $\cos\theta$.

Y coordinate of P is defined as $\sin\theta$.

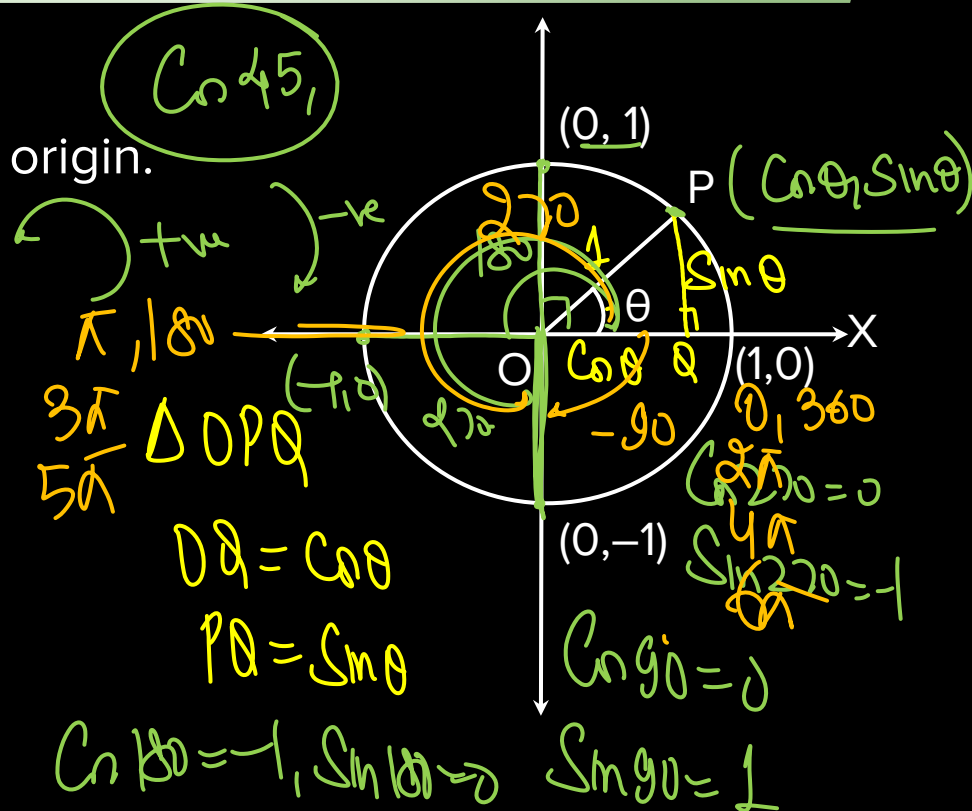
360 / radians

2π = 360

$\sin\theta = \frac{PQ}{r}$

$\cos\theta = \frac{OQ}{r}$

1 rad = 180°
π/2 = 90°



$$n \in \mathbb{I}$$

Observations:

1. (a) $\sin n\pi = 0$

$$\sin(180n) = 0$$

(b) $\cos 2n\pi = 1$ ✓

$$\cos(180 \cdot 2n) = 1$$

(c) $\cos(2n+1)\pi = -1$

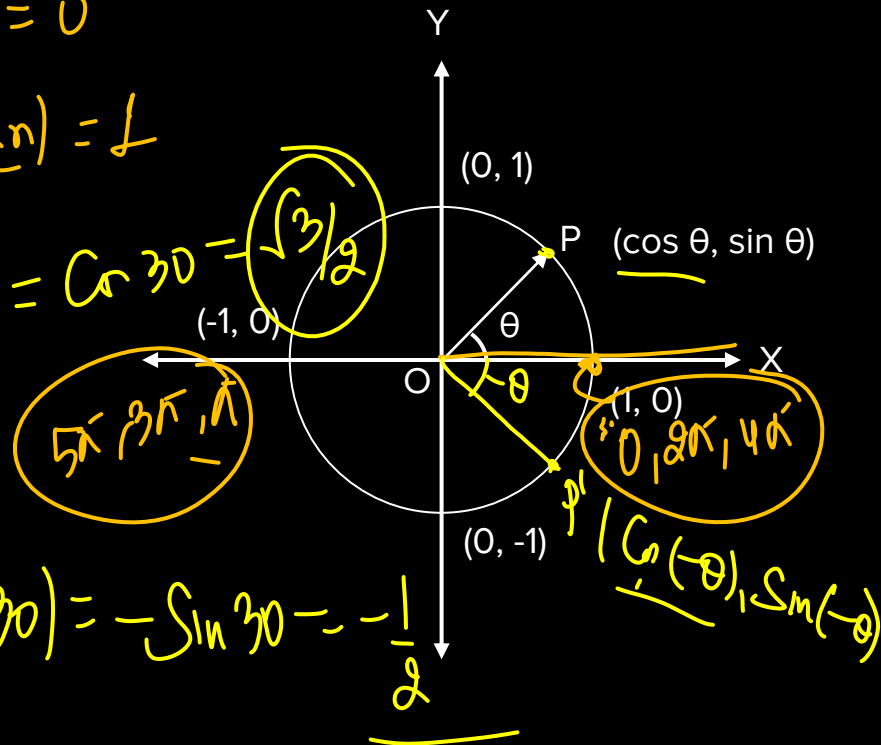
$$\cos(-30) = \cos 30 = \frac{\sqrt{3}}{2}$$

2. (a) $\cos(-\theta) = \cos \theta$

(b) $\sin(-\theta) = -\sin \theta$

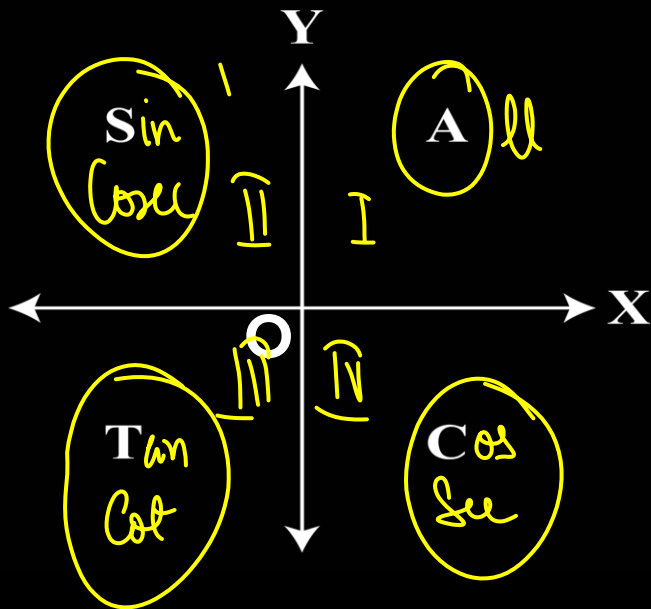
(c) $\tan(-\theta) = -\tan \theta$

$$\sin(-30) = -\sin 30 = -\frac{1}{2}$$



Observations:

Lets see the signs of trigonometric ratios in different quadrants



After School to College
All Sine to Cos

Observations:

1. Sine of supplementary angles are same.
2. Cosines of supplementary angles are negative of each other.
3. $\sin(-\theta) = -\sin\theta$ and $\cos(-\theta) = \cos\theta$

$$\checkmark \checkmark \cos 170 = -\cos 10$$

$$\cos 165 = -\cos 15$$

$$\theta, 180-\theta$$

$$\boxed{\sin(180-\theta) = \sin\theta}$$

$\checkmark \checkmark$

$$\sin 170 = \sin 10$$

$$\sin 165 = \sin 15$$

Compound Angles Formulae

$$1. \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$2. \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$3. \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$4. \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$5. \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$6. \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$7. \cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$$

$$8. \cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

Optional for
for RMO $\sin 75^\circ$
 $\cos 75^\circ$

$$\sin 15^\circ = \sin(45^\circ - 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ$$

Multiple / Submultiple Angle Formulae

1. $\sin 2\theta = 2 \sin \theta \cos \theta$

2. $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

3. $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

Geometry

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$A = B = \theta$

$1 - c^2$

$\cos 2\theta$

$\cos^2 \theta - \sin^2 \theta$

$2\cos^2 \theta - 1$

$1 - 2\sin^2 \theta$

$1 + \cos 2\theta = 2 \cos^2 \theta$

$1 - \cos 2\theta = 2 \sin^2 \theta$

Trigonometric formulae

$\frac{1}{2} (2 \sin \theta \cos \theta)$

$= \frac{1}{2} \sin 2\theta$

Multiple / Submultiple Angle Formulae

Remark:

a. $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$

b. $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$

c. $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

Compulsory

optional

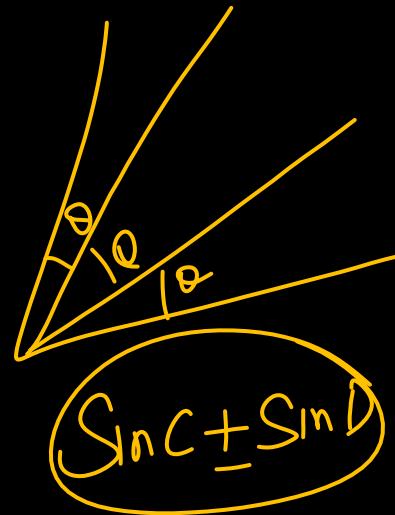
for the RMO

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= 2 \cos^2 \theta - 1$$

$$= 1 - 2 \sin^2 \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$



$$\sqrt{x^2} = |x|$$

Q1. Simplify the expression

$$\sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}$$

A $\cos 2x$

B $\cos 3x$

C $\sin 3x$

D $\sin 2x$

$$\begin{aligned} & \sqrt{S^4 + 4(1-S^2)} = |S^2 - 2| \\ & = \sqrt{S^4 + 4 - 4S^2} = (\cancel{2} - S^2) - (\cancel{2} - C^2) \\ & = \sqrt{(S^2)^2 + 2^2 - 2 \cdot 2S^2} = C^2 - S^2 \\ & = \sqrt{(S^2 - 2)^2} = \boxed{\cos 2x} \end{aligned}$$

Q1. Simplify the expression

$$\sqrt{\sin^4 x + 4\cos^2 x} - \sqrt{\cos^4 x + 4\sin^2 x}.$$

A $\cos 2x$

B $\cos 3x$

C $\sin 3x$

D $\sin 2x$

Solution:

The given expression is equal to

$$\begin{aligned}& \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)} \\&= \sqrt{(2 - \sin^2 x)^2} - \sqrt{(2 - \cos^2 x)^2} \\&= (2 - \sin^2 x) - (2 - \cos^2 x) \\&= \cos^2 x - \sin^2 x = \cos 2x.\end{aligned}$$

$$k = 9$$

Obs. (OP)

Q2. If $(4\cos^2 9^\circ - 3)(4\cos^2 27^\circ - 3) = \tan k^\circ$. Find the value of k

$$4\cos^2 9 - 3 = \frac{\cos 9(4\cos^2 9 - 3)}{\cos 9} = \frac{(4\cos^3 9 - 3\cos 9)}{\cos 9}$$

$$4\cos^3 \theta - 3\cos \theta = \cos 3\theta$$

$$\sin \theta = \cos(90^\circ - \theta)$$

$$\frac{\cancel{\cos 27}}{\cos 9} \cdot \frac{\cos 81}{\cancel{\cos 27}}$$

$$\begin{aligned} \sin 9 &= \frac{\cos 27}{\cos 9} \\ &= \frac{\cos 81}{\cos 9} = \tan 9 \end{aligned}$$

Q2. If $(4\cos^2 9^\circ - 3)(4\cos^2 27^\circ - 3) = \tan k^\circ$. Find the value of k

Answer: 9

Solution:

We have $\cos 3x = 4\cos^3 x - 3\cos x$,

So $4\cos^2 x - 3 = \frac{\cos 3x}{\cos x}$ for all $x \neq (2k + 1) \cdot 90^\circ, k \in \mathbb{Z}$.

Thus

$$(4\cos^2 9^\circ - 3)(4\cos^2 27^\circ - 3) = \frac{\cos 27^\circ}{\cos 9^\circ} \cdot \frac{\cos 81^\circ}{\cos 27^\circ} = \frac{\cos 81^\circ}{\cos 9^\circ}$$

$$= \frac{\sin 9^\circ}{\cos 9^\circ} = \tan 9^\circ,$$

as desired.

Q3. Find the smallest positive integer n such that

$$\frac{\sin 1^\circ}{\sin 45^\circ \sin 46^\circ} + \frac{\sin 1^\circ}{\sin 47^\circ \sin 48^\circ} + \dots + \frac{\sin 1^\circ}{\sin 133^\circ \sin 134^\circ} = \frac{\sin 1^\circ}{\sin n^\circ}$$

$$\frac{\sin 46^\circ \cos 45^\circ - \cos 46^\circ \sin 45^\circ}{\sin 45^\circ \sin 46^\circ}$$

$$= \frac{\sin 46^\circ \cos 45^\circ}{\sin 45^\circ \sin 46^\circ} - \frac{\cos 46^\circ \sin 45^\circ}{\sin 45^\circ \sin 46^\circ}$$

$$= (\cot 45^\circ - \cot 46^\circ)$$

$$[\cot 45^\circ - \cot 46^\circ] + [\cot 47^\circ - \cot 48^\circ] + [\cot 49^\circ - \cot 50^\circ] + [\cot 51^\circ - \cot 52^\circ] = \frac{\sin 1^\circ}{\sin n}$$

$$+ \dots + [\cot 133^\circ - \cot 134^\circ]$$

$$\cot 45^\circ - \cot 90^\circ = \frac{\sin 1^\circ}{\sin n}$$

$$\sin n = \sin 1^\circ$$

$$n = 1$$

$$46 + 134 = 180$$

$$\cot(134) = -\cot 46$$

$$\cot 134 + \cot 46 = 0$$

$$134 - 45 + 1$$

$$135 - 45$$

$$= 90$$

$$\sin, \cos$$

Q3. Find the smallest positive integer n such that

$$\frac{1}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^\circ \sin 48^\circ} + \dots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \frac{1}{\sin n^\circ}$$

Answer: 1

Solution:

Multiplying both sides of the given equation by $\sin 1^\circ$, we have

$$\frac{\sin 1^\circ}{\sin n^\circ}$$

$$\begin{aligned} &= (\cot 45^\circ - \cot 46^\circ) + (\cot 47^\circ - \cot 48^\circ) + \dots + (\cot 133^\circ - \cot 134^\circ) \\ &= \cot 45^\circ - (\cot 46^\circ + \cot 134^\circ) + (\cot 47^\circ + \cot 133^\circ) - \dots + (\cot 89^\circ + \cot 91^\circ) - \cot 90^\circ \\ &= 1. \end{aligned}$$

Therefore, $\sin n^\circ = \sin 1^\circ$, and the least possible integer value for n is 1.

Coef $\rightarrow$ exchange (55)

AMC 12 1999

JEE

Q4. In triangle ABC,

$3\sin A + 4\cos B = 6$ and $4\sin B + 3\cos A = 1$. Find the measure of angle C in degrees.

$3\sin A + 4\cos B$ — max

$A + B = 30^\circ, B = 0^\circ, A = 30^\circ$

$(A+B) + C = 180^\circ$

$$\begin{aligned} 3\sin A + 4\cos B &= 6 \\ 4\sin B + 3\cos A &= 1 \end{aligned}$$

S_2 & adding

$\frac{3}{2} + 4 \leq 1.5 + 4$

$$9\sin^2 A + 16\cos^2 B + 24\sin A \cos B = 36$$

$$9\cos^2 A + 16\sin^2 B + 24\sin B \cos A = 1$$

$$9 + 16 + 24\sin(A+B) = 37$$

$$\sin(A+B) = \frac{1}{2}$$

$\sin C = \frac{1}{2}$

$C \rightarrow 30^\circ$

~~150°~~

~~$\text{If } C \rightarrow 150^\circ, A+B = 30^\circ$~~

Q4. In triangle ABC,
 $3\sin A + 4\cos B = 6$ and $4\sin B + 3\cos A = 1$. Find the measure
of angle C in degrees.

Answer: 30°

Solution:

Square the two given equations and add the results to obtain

$$24 (\sin A \cos B + \cos A \sin B) = 12,$$

or $\sin(A + B) = \frac{1}{2}$. Because $C = 180^\circ - A - B$,

we have $\sin C = \sin(A + B) = \frac{1}{2}$,

Implying that either $C = 30^\circ$ or $C = 150^\circ$.

But if $C = 150^\circ$, $A < 30^\circ$, so $3 \sin A + 4 \cos B < \frac{3}{2} + 4 < 6$, a contradiction.

Hence the answer is $C = 30^\circ$.

Q5. Given that $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, find n .

Hint

$$A + B = 45$$

$$1 + \tan A + \tan B + \tan A \tan B$$

$$(1 + \tan A)(1 + \tan B) = 2$$

$$A + B = 45$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan A + \tan B = -\tan A + B$$

$$\tan A + \tan B + \tan A \tan B = 1$$

2^n ,
 $f(1+\tan 1) \cdot (1+\tan 2)$
 $(1+\tan 3) \cdot$
 $(1+\tan 4) \cdot (1+\tan 5)$
 $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$
 (2^6)

2^{23}

Q5. Given that $(1 + \tan 1^\circ) (1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, find n .

Answer: 23

Solution:

$$\begin{aligned}& (1 + \tan k^\circ) (1 + \tan(45 - k)^\circ) \\&= 1 + [\tan k^\circ + \tan (45 - k)^\circ] + \tan k^\circ \tan (45 - k)^\circ \\&= 1 + \tan 45^\circ [1 - \tan k^\circ \tan (45 - k)^\circ] + \tan k^\circ \tan (45 - k)^\circ \\&= 2.\end{aligned}$$

Hence,

$$\begin{aligned}& (1 + \tan 1^\circ) (1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) \\&= (1 + \tan 1^\circ) (1 + \tan 44^\circ) (1 + \tan 2^\circ) (1 + \tan 43^\circ) \dots (1 + \tan 22^\circ) (1 + \tan 23^\circ) (1 + \tan 45^\circ) \\&= 2^{23},\end{aligned}$$

implying that $n = 23$.

Q6. If $x = \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 89^\circ$ and
 $y = \cos 2^\circ \cos 6^\circ \cos 10^\circ \dots \cos 86^\circ$, then what is the
integer nearest to $\frac{2}{7} \log_2 \left(\frac{y}{x} \right)$?

A 13

B 15

C 17

D 19

Solution:

$$\begin{aligned}\frac{y}{x} &= \frac{\cos 2^\circ \cos 6^\circ \cos 10^\circ \dots \cos 86^\circ}{\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 89^\circ} \\&= 2^{44} \times \sqrt{2} \frac{\cos 2^\circ \cos 6^\circ \cos 10^\circ \dots \cos 86^\circ}{\sin 2^\circ \sin 4^\circ \sin 6^\circ \dots \sin 88^\circ} \\&= \frac{2^{\frac{89}{2}} \sin 4^\circ \sin 8^\circ \sin 12^\circ \dots \sin 88^\circ}{\sin 2^\circ \sin 4^\circ \sin 6^\circ \dots \sin 88^\circ} \\&= \frac{2^{\frac{89}{2}}}{2^{\frac{89}{2}}} \\&= \frac{\cos 4^\circ \cos 8^\circ \cos 12^\circ \dots \cos 88^\circ}{\cos 4^\circ \cos 8^\circ \cos 12^\circ \dots \cos 88^\circ} \\&= \frac{2^{\frac{89}{2}}}{1} = 2^{\frac{89}{2} + 22} \\&= \frac{1}{2^{22}}\end{aligned}$$

$$= 2^{\frac{133}{2}}$$

$$\Rightarrow \frac{2}{7} \log_2 \left(\frac{y}{x} \right) = \frac{2}{7} \log_2 2^{\frac{133}{2}}$$

$$= \frac{2}{7} \times \frac{133}{2} = 19$$

NIGERT-11

Example 11

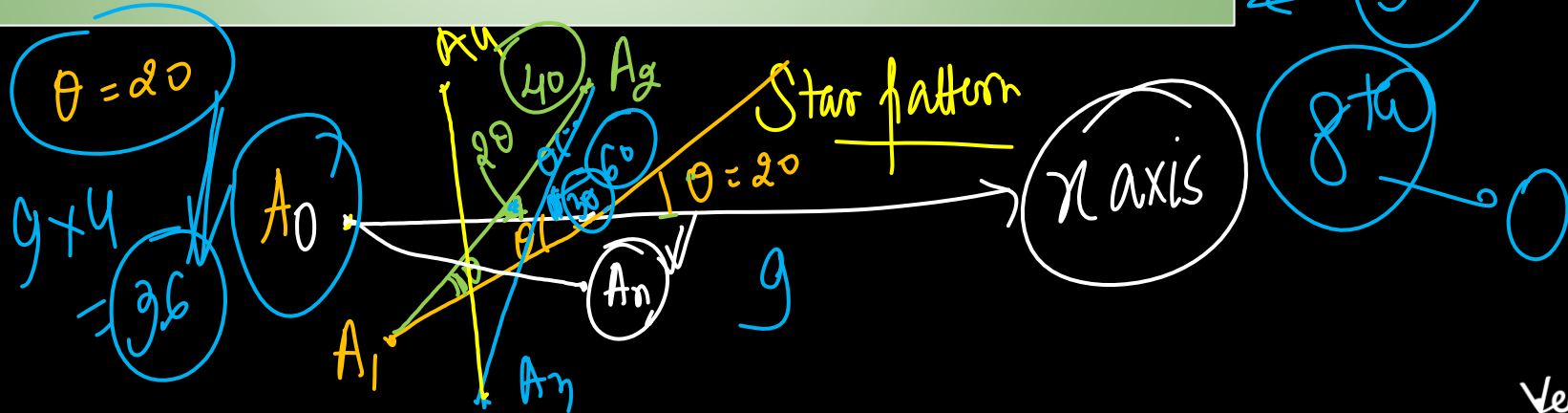
R.D. Sharma

11

Q7. An ant leaves the anthill for its morning exercise. It walks 4 feet east and then makes a 160° turn to the right and walks 4 more feet. It then makes another 160° turn to the right and walks 4 more feet. If the ant continues this pattern until it reaches the anthill again, what is the distance in feet it would have walked?

Last Question

20, 40, 60, 80



Q7. An ant leaves the anthill for its morning exercise. It walks 4 feet east and then makes a 160° turn to the right and walks 4 more feet. It then makes another 160° turn to the right and walks 4 more feet. If the ant continues this pattern until it reaches the anthill again, what is the distance in feet it would have walked?

Answer: 36 feet

Solution:

Let $A_0(0, 0)$

$A_1(4 \cos 0^\circ, 4 \sin 0^\circ)$

$A_2(4 \cos 0^\circ + 4 \cos 160^\circ, 4 \sin 0^\circ + 4 \sin 160^\circ)$

$A_n = (0, 0)$

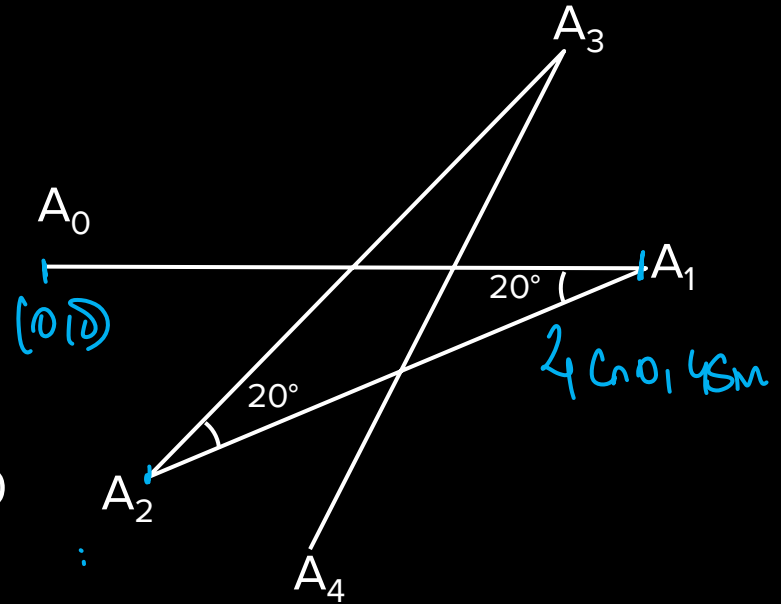
$$\Rightarrow 4(\cos 0^\circ + \cos 160^\circ + \dots + \cos 160^\circ(n-1)) = 0$$

$$\text{and } 4(\sin 0^\circ + \sin 160^\circ + \dots + \sin 160^\circ(n-1)) = 0$$

$$\Rightarrow \sin(80^\circ n) = 0$$

$$\Rightarrow n = 9$$

$$\Rightarrow \text{distance covered} = 4 \times 9 = 36 \text{ feet}$$





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Power of point

$\sin(A \pm B)$
 $\cos(A \pm B)$

$\sin 30^\circ$
 $\cos 30^\circ$
 $\sin 45^\circ$
 $\cos 45^\circ$
 $\sin 60^\circ$
 $\cos 60^\circ$

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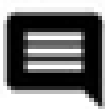
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